

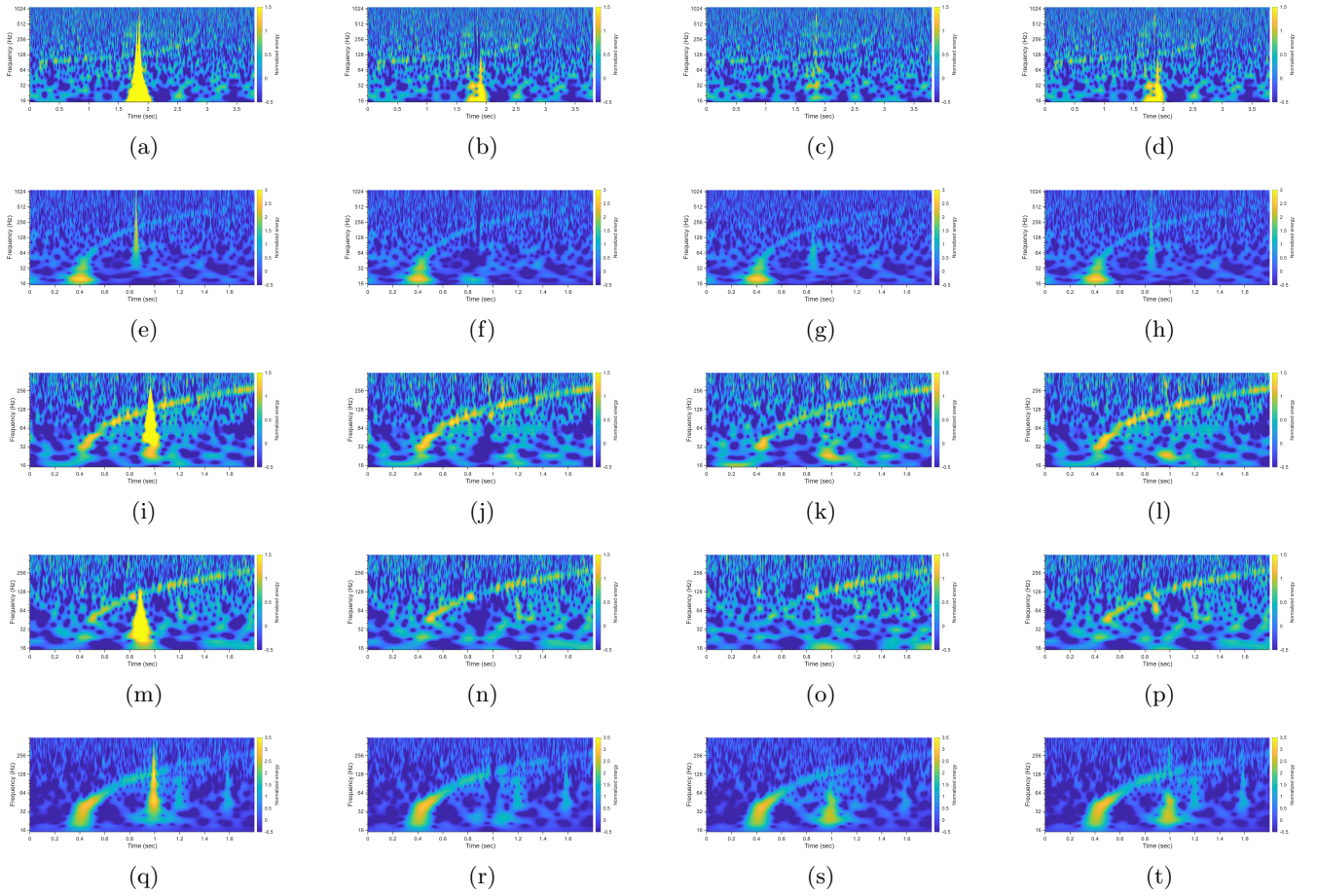


FIG. 8: Q-transform comparison of subtraction technique across five glitch instances: GW170817, Blip, Koi Fish, Tomte, and low frequency blip, from top to bottom. Each row shows, from left to right, the raw whitened data, the SHAPES residual, the Waveshrink (WS) residual, and the SHAPES+WS residual, all on the same time window.

The segment boundary used to build every residual in a row comes from FLARE, held fixed across all three subtraction techniques so that differences between panels reflect the subtraction technique rather than the identification method. Each row uses its own colour scale, chosen to match the power range of that glitch, since a single scale shared across all five rows would suppress the visibility of the lower-power low frequency blip relative to louder cases such as GW170817 and Blip. GW170817 carries no injected chirp and is included for qualitative comparison of residual structure only; the signal-to-noise recovery reported in Table II applies to the four real glitch instances beneath it, into which a chirp signal of known parameters was injected to test subtraction performance.

tens of seconds per glitch, and its cost tracks the boundary it is given rather than the identification method's own runtime: averaged over the four injected glitches, SHAPES needs about 52.8 s under AMPS boundaries, 49.2 s under CRISP, and only 26.5 s under FLARE, even though FLARE's own identification stage is the slowest of the three by a wide margin. GW170817 costs SHAPES the most fitting time under every method, 113.95 s under AMPS, 70.34 s under FLARE, and 51.37 s under CRISP, consistent with its wider, more complex boundary. Combined's total cost is SHAPES's time plus a WS pass adding a few thousandths of a second, so the small SNR gain Combined holds over SHAPES alone comes at negligible extra cost.

VI. CONCLUSION

We presented three glitch boundary identification methods, AMPS, FLARE, and CRISP, and tested each against three subtraction techniques, SHAPES, wavelet shrinkage, and their combination, across five glitch instances drawn from three observing runs. AMPS relies only on a robust amplitude threshold applied to the whitened strain. FLARE builds its boundary from the best fitness of a spline model fit segment by segment. CRISP localizes a glitch from spectrogram power alone and uses AMPS only to confirm a candidate inside a boundary it has already set.

All three recover an injected chirp signal with a residual SNR close to the injected value for the four broad-

band glitch cases, and CRISP gives the most consistent boundary width across glitches of markedly different duration, settling near 0.11 s for four of the five instances tested. AMPS remains the fastest identification method by a wide margin, completing in well under a tenth of a second per glitch against tens of seconds for FLARE, while CRISP sits a small factor above AMPS without approaching FLARE’s cost.

The subtraction comparison across the three identification methods points to the same ranking. SHAPES combined with wavelet shrinkage gives the highest recovered SNR for Blip, Tomte, and Koi Fish under all three boundary methods, recovering 95 to 97 percent of the injected value on average, and wavelet shrinkage alone trails both SHAPES and the combined technique by a wide margin in every case tested, recovering only 64 percent. The one departure from this pattern is the low frequency blip, where SHAPES alone edges out the combined technique under AMPS, though the two stay close. This narrow departure does not track the broader recovery pattern, in which the low frequency blip performs among the best of the four glitches rather than the worst, so it is best read as a small, boundary-specific fluctuation rather than a consequence of the glitch’s own frequency content.

Injection position within the overlap window affects recovered SNR, and Appendix C shows that Koi Fish’s recovered SNR under SHAPES and Combined can exceed the injected value at some positions, a chance effect of the noise realization rather than a property of subtraction [56, 57]. The present study is limited to five glitch instances. Other glitch morphologies, including scattered light, are not yet exercised. The threshold values used throughout this work, including the amplitude threshold in AMPS, the fitness threshold in FLARE, and the two power-mask fractions in CRISP, were each set from a single glitch instance or a small handful of them. These values should be solidified against a much larger and more diverse sample before they can be treated as defaults rather than as instance-specific settings. This scale of evaluation is consistent with prior model-based glitch-subtraction studies [11, 12, 59], though unlike SHAPES’s single parameter, whose exact value was found to matter little, the threshold values here were tuned directly against the instances used for evaluation, a stronger dependency worth flagging separately. Building that sample by hand is impractical, since instances of the same morphological class carry substantially different glitch SNR and background conditions even within one class. A machine-learning-based parameter search across a larger, automatically labeled sample is a natural next step for fixing threshold values that currently rest on a small number of hand-checked instances.

Future work should extend the comparison across the full range of GravitySpy glitch classes, with particular attention to scattered light and other narrowband or slowly varying morphologies, since the amplitude and periodicity criteria built into CRISP for this purpose have not

yet been tested against labelled scattering data. One planned extension is a second, lower amplitude threshold applied to a filtered version of the bounded segment, aimed at separating scattering glitches from the transient classes studied here, since scattering typically shows a weaker and more spread-out amplitude signature than a genuine transient. This criterion is not yet validated and remains untested in the current results. Recent work on adaptive, template-free subtraction of scattering arches suggests one path forward for this extension [26]. The interaction between glitch frequency content and the injected signal, which affected recovery differently across the four glitch classes in this work, also warrants a systematic study across a range of injected signal frequencies rather than the single chirp profile used here. Finally, this work tests only one real overlapping signal, GW170817, and relies on a synthetic linear chirp for the remaining four glitches. Extending the comparison to a larger sample of real compact-binary signals, spanning a range of masses and overlap geometries, would test whether the present conclusions hold beyond the single real case and the simplified injection used here as a stand-in. Whether the position-dependence identified in Appendix C propagates to biases in estimated source parameters for a real overlapping signal, rather than only to recovered SNR, is a separate question this work does not address.

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Appendix A: Matched-filter check for the GW170817 instance

Section IV treats the GW170817 instance as a qualitative check rather than a quantitative recovery test, since no signal was injected for this case. As a supplementary check, a matched-filter test was carried out on the SHAPES, wavelet-shrinkage, and combined residuals obtained from the CRISP-bounded segment. The template is a Newtonian, restricted-post-Newtonian chirp with component masses of 1.46 and 1.30 solar masses, close to the values reported for GW170817. A sliding matched filter was applied across each residual, and the peak of the resulting signal-to-noise ratio time series was recorded, following the same convention used elsewhere in this work for the injected-chirp cases.

The data segment used for this check spans 8 s, shorter than the segment needed to hold a chirp starting at the

conventional 40 Hz low-frequency cutoff. The template’s starting frequency was therefore raised to approximately 63 Hz so that the full chirp fits within the available segment. This choice trades template duration for consistency with the segment length already used for boundary identification in this work, and the resulting statistic should be read as an internal, self-consistent check across the three residuals rather than a value comparable to matched-filter statistics computed over the full inspiral band.

Figure 9 shows the envelope of the matched-filter SNR time series for the three residuals, restricted to a narrow window around the shared peak. All three residuals show a clear, sharply peaked maximum at the same arrival time, confirming that the real signal survives subtraction under all three techniques, though wavelet shrinkage alone recovers less of it. The peak SNR is 8.20 for SHAPES alone, 6.66 for wavelet shrinkage alone, and 8.20 for the combined technique. SHAPES alone and the combined technique track one another closely across the entire window, consistent with their near-identical peak values, while wavelet shrinkage alone sits visibly below both throughout, not only at the peak. This pattern matches the identification-and-subtraction comparison in the main text, where wavelet shrinkage alone consistently trails SHAPES and SHAPES with wavelet shrinkage combined.

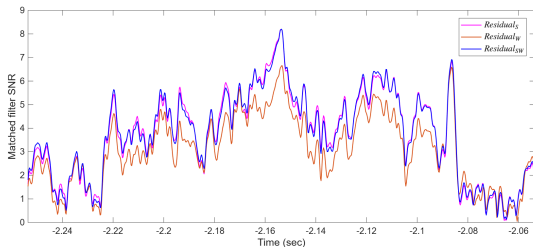


FIG. 9: Envelope of the matched-filter signal-to-noise ratio for the CRISP-bounded GW170817 residual, shown for SHAPES alone (magenta), wavelet shrinkage alone (orange), and the combined technique (blue). The horizontal axis is time in seconds relative to the matched-filter lag, and the vertical axis is the matched-filter signal-to-noise ratio. Wavelet shrinkage alone recovers visibly less signal than the other two techniques across the shown window, while SHAPES alone and the combined technique remain close to one another throughout.

Appendix B: Sensitivity of the boundary to the threshold multiplier

The boundary used by CRISP and FLARE in Section V is set from a power threshold applied to the spectrogram columns of the candidate segment, scaled by a fixed multiplier before it is compared against the en-

velope described in Section II. In Section V this multiplier is fixed at 3.0 for every glitch instance, which keeps the comparison across the five cases consistent, but leaves open how much residual signal power depends on that particular choice. We examine this question for GW170817, the one instance in our sample where the underlying glitch carries a low GravitySpy machine-learning confidence despite its high reported loudness, and where the default multiplier leaves a visible patch of unsubtracted power at low frequency.

Lowering the multiplier from 3.0 to 1.5 widens the boundary from 448 to 640 samples, or from 0.1094 s to 0.1562 s (Table IV), because a smaller multiplier lowers the power level a spectrogram column must clear to be kept inside the boundary, so more of the glitch’s lower-power edges are retained. This widening tracks directly with subtraction quality: Figure 10 shows that the low-frequency patch left below 40 Hz after subtraction at the default multiplier shrinks steadily as the multiplier is lowered toward 1.5, for both SHAPES and SHAPES+WS, while the broadband content of the glitch above roughly 60 Hz is already removed cleanly at every multiplier tested. We did not find a multiplier within this range that removes the low-frequency patch entirely.

TABLE IV: Boundary obtained for the glitch overlapping GW170817 as the threshold multiplier is varied. A smaller multiplier lowers the power level required to keep a spectrogram column inside the boundary, which widens the resulting segment.

Multiplier(m)	Threshold	Samples	Boundary (s)
3.0	0.60	448	0.1094
2.5	0.50	512	0.1250
2.0	0.40	576	0.1406
1.5	0.30	640	0.1562

A wider boundary is not free of cost. It also lets more of the surrounding background into the fitting segment, which was the original reason for keeping the boundary tight, so the trend in Table IV and Figure 10 should not be read as evidence that a smaller multiplier is a better general-purpose choice, only that this particular glitch’s low-frequency residual is sensitive to it.

We attribute the difficulty of this case to the ambiguous morphological status of the GW170817 glitch itself, rather than to a shortcoming specific to CRISP or FLARE. GravitySpy reports high loudness for this glitch but low classifier confidence, meaning the automated pipeline that labels glitch morphology is itself uncertain what class this glitch belongs to. A boundary-identification method tuned and tested against clearly labeled Blip, Tomte, and KoiFish instances has, by construction, less reason to generalize cleanly to a glitch that resists confident classification.

This result reinforces a point already raised in the conclusion. The threshold values used throughout this work, including the multiplier examined here, were set from a

small number of instances and should be solidified against a larger and more varied sample before being treated as defaults. We leave a systematic study of the multiplier across the full range of glitch classes, together with the scattering case, as future work.

A comparable residual appears in earlier work on this same glitch. Mohanty and Chowdhury [12] split the GW170817 segment into three separately tuned pieces to manage its extended low-frequency wings, and reported residual overfitting below 32 Hz that they judged acceptable since it did not overlap the signal band. The residual we report here under a single, automatically determined boundary is consistent with that earlier finding: closing it further would require a smaller multiplier and shorter segment length, which risks removing signal power along with the glitch, so we accept the residual under the default setting rather than force complete removal.

Appendix C: Sensitivity of Koi Fish recovery to injection position

The injected chirp for each glitch instance in Section IV begins at a fixed onset time t_a relative to the glitch, chosen so the injection window overlaps the glitch for its full duration. For Koi Fish, this onset was varied across three values, $t_a = 0.8, 0.5$, and 0.3 s, holding the injected SNR target, injection window length, and frequency sweep fixed, to test how sensitive recovered SNR is to this choice and to check whether the position used in the main text, $t_a = 0.5$ s, is representative rather than a favorable outlier.

All three identification methods and three subtraction techniques were rerun on Koi Fish at each position, following the same pipeline used throughout Section V. Table V reports the recovered SNR at each position.

Recovered SNR for SHAPES and Combined rises steadily as t_a increases from 0.3 to 0.8 s, and at $t_a = 0.8$ s exceeds the injected value of 30 under all three identification methods, by 0.2 to 3.5 percent depending on the method and subtraction technique. WS alone follows the same trend but stays below the injected value at every position tested. At $t_a = 0.5$ s, the position used in Section V, SHAPES and Combined both recover between about 89 and 99 percent of the injected value, with no case exceeding 100 percent.

This pattern is consistent with the excess at $t_a = 0.8$ s arising from a chance contribution of the noise realization entering the matched filter at that specific position, rather than from a property of SHAPES or the combined technique. Under stationary, Gaussian noise the observed matched-filter SNR is distributed normally around the true optimal SNR with unit variance [57], so a residual can register a slightly higher SNR than injected purely because the noise segment sampled at a given position happens to correlate with the template. Hardware injection studies at LIGO report deviations between recovered and expected SNR of around twenty percent from

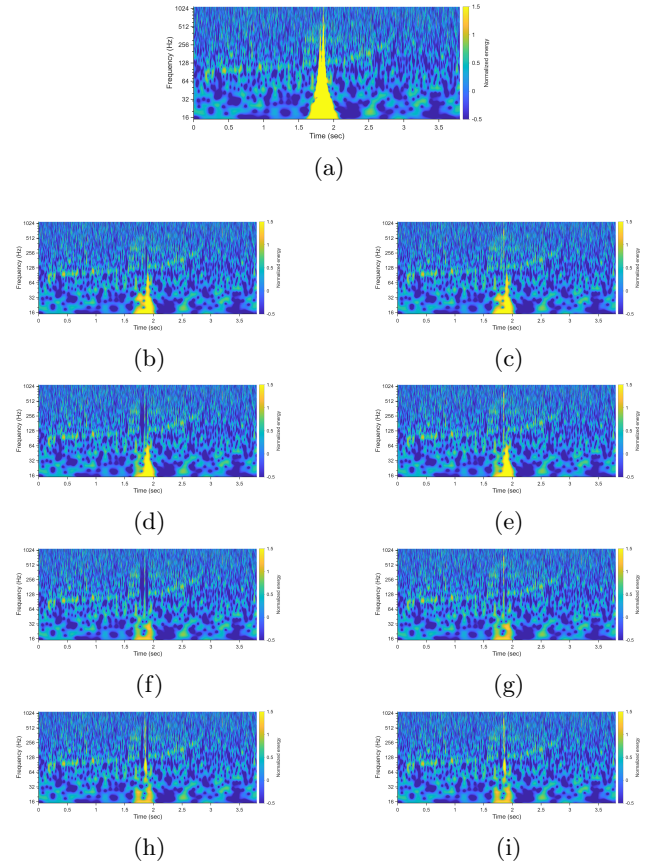


FIG. 10: Effect of the boundary threshold multiplier on subtraction of the glitch overlapping GW170817. Top: Q-transform of the glitch before subtraction. Below, each row corresponds to one multiplier, from top to bottom: 3.0 (default), 2.5 , 2.0 , and 1.5 . Left column: Q-transform of the SHAPES residual at the corresponding multiplier. Right column: Q-transform of the SHAPES+WS residual at the corresponding multiplier. In every panel, the horizontal axis is time in seconds and the vertical axis is frequency in Hz, plotted on a logarithmic scale. Color indicates normalized energy, on a logarithmic scale, and the colorbar range is shared across all panels, including the reference panel at top. The low-frequency patch below 40 Hz shrinks as the multiplier decreases from 3.0 to 1.5 , while the broadband high-frequency content of the glitch is removed at every multiplier shown.

this and related calibration effects [56], consistent with the size of the excess seen here at $t_a = 0.8$ s.

WS alone shows a separate effect not reducible to this explanation. Its recovered SNR drops by close to three points between $t_a = 0.8$ s and $t_a = 0.5$ s or 0.3 s, a larger proportional shift than seen for SHAPES or Combined at the same positions. Since WS acts directly on the residual without a prior spline fit, this suggests recovered SNR under WS alone is more sensitive to where the injected chirp sits relative to the glitch than the other

two techniques, independent of the specific excess seen at $t_a = 0.8$ s.

A three-point sweep at one glitch instance does not establish how recovered SNR depends on injection position in general, and a broader study across positions and glitch classes would be needed for that. Whether this position-dependence propagates to biases in estimated source parameters for a real overlapping signal, rather than only to recovered SNR, is also not addressed here. We leave both as future work, together with the broader parameter study proposed in Section VI.

TABLE V: Recovered signal-to-noise ratio for Koi Fish at three injection onset times t_a , compared across the three identification methods (AMPS, FLARE, CRISP) and three subtraction techniques (SHAPES, WS, Combined). The injected SNR target is 30 at every position. $t_a = 0.5$ s is the position used in Section V.

Method	SHAPES	WS	Combined
$t_a = 0.8$ s			
AMPS	30.0686	22.1094	30.3025
FLARE	31.0567	22.1125	31.0063
CRISP	29.8846	22.1094	30.0330
$t_a = 0.5$ s (<i>used in Section V</i>)			
AMPS	29.2630	19.4828	29.4823
FLARE	29.8092	19.4886	29.7993
CRISP	29.6471	19.4828	29.7679
$t_a = 0.3$ s			
AMPS	27.6340	19.7183	27.7264
FLARE	28.5323	19.7201	28.6765
CRISP	27.3349	19.7183	27.5045

[1] B. P. Abbott, R. Abbott, T. D. Abbott, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, R. X. Adhikari, V. B. Adya, C. Affeldt, M. Afrough, B. Agarwal, M. Agathos, K. Agatsuma, N. Aggarwal, O. D. Aguiar, L. Aiello, A. Ain, P. Ajith, B. Allen, G. Allen, A. Allocca, P. A. Altin, A. Amato, A. Ananyeva, S. B. Anderson, W. G. Anderson, S. V. Angelova, S. Antier, S. Appert, K. Arai, M. C. Araya, J. S. Areeda, N. Arnaud, K. G. Arun, S. Ascenzi, G. Ashton, M. Ast, S. M. Aston, P. Astone, D. V. Atallah, P. Aufmuth, C. Aulbert, K. AultONeal, C. Austin, A. Avila-Alvarez, S. Babak, P. Bacon, M. K. M. Bader, S. Bae, M. Bailes, P. T. Baker, F. Baldaccini, G. Ballardin, S. W. Ballmer, S. Banagiri, J. C. Barayoga, S. E. Barclay, B. C. Barish, D. Barker, K. Barkett, F. Barone, B. Barr, L. Barsotti, M. Barsuglia, D. Barta, S. D. Barthelmy, J. Bartlett, I. Bartos, R. Bassiri, A. Basti, J. C. Batch, M. Bawaj, J. C. Bayley, M. Bazzan, B. Bécsy, C. Beer, M. Bejger, I. Belahcene, A. S. Bell, B. K. Berger, G. Bergmann, S. Bernuzzi, J. J. Bero, C. P. L. Berry, D. Bersanetti, A. Bertolini, J. Betzwieser, S. Bhagwat, R. Bhandare,

I. A. Bilenko, G. Billingsley, C. R. Billman, J. Birch, R. Birney, O. Birnholtz, S. Biscans, S. Biscoveanu, A. Bisht, M. Bitossi, C. Biwer, M. A. Bizouard, J. K. Blackburn, J. Blackman, C. D. Blair, D. G. Blair, R. M. Blair, S. Bloemen, O. Bock, N. Bode, M. Boer, G. Bogaert, A. Bohe, F. Bondu, E. Bonilla, R. Bonnand, B. A. Boom, R. Bork, V. Boschi, S. Bose, K. Bossie, Y. Bouffanais, A. Bozzi, C. Bradaschia, P. R. Brady, M. Branchesi, J. E. Brau, T. Briant, A. Brillet, M. Brinkmann, V. Brisson, P. Brockill, J. E. Broida, A. F. Brooks, D. A. Brown, D. D. Brown, S. Brunett, C. C. Buchanan, A. Buikema, T. Bulik, H. J. Bulten, A. Buonanno, D. Buskulic, C. Buy, R. L. Byer, M. Cabero, L. Cadonati, G. Cagnoli, C. Cahillane, J. Calderón Bustillo, T. A. Callister, E. Calloni, J. B. Camp, M. Canepa, P. Canizares, K. C. Cannon, H. Cao, J. Cao, C. D. Capano, E. Capocasa, F. Carbognani, S. Caride, M. F. Carney, G. Carullo, J. Casanueva Diaz, C. Casentini, S. Caudill, M. Cavaglià, F. Cavalier, R. Cavalieri, G. Cella, C. B. Cepeda, P. Cerdá-Durán, G. Cerretani, E. Cesarini, S. J. Chamberlin, M. Chan, S. Chao,

- P. Charlton, E. Chase, E. Chassande-Mottin, D. Chatterjee, K. Chatziioannou, B. D. Cheeseboro, H. Y. Chen, X. Chen, Y. Chen, H.-P. Cheng, H. Chia, A. Chincarini, A. Chiummo, T. Chmiel, H. S. Cho, M. Cho, J. H. Chow, N. Christensen, Q. Chu, A. J. K. Chua, and S. Chua, GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral, *Phys. Rev. Lett.* **119**, 161101 (2017), arXiv:1710.05832 [gr-qc].
- [2] B. P. Abbott, R. Abbott, T. D. Abbott, M. R. Abernathy, F. Acernese, K. Ackley, C. Adams, T. Adams, and A. et al., Effects of data quality vetoes on a search for compact binary coalescences in Advanced LIGO's first observing run, *Classical and Quantum Gravity* **35**, 065010 (2018), arXiv:1710.02185 [gr-qc].
- [3] R. Girgaonkar and S. D. Mohanty, Glitch veto based on unphysical gravitational wave binary inspiral templates, *Phys. Rev. D* **110**, 023037 (2024), arXiv:2401.15237 [gr-qc].
- [4] S. A. Usman, A. H. Nitz, I. W. Harry, C. M. Biwer, D. A. Brown, M. Cabero, C. D. Capano, T. Dal Canton, T. Dent, S. Fairhurst, M. S. Kehl, D. Keppel, B. Krishnan, A. Lenon, A. Lundgren, A. B. Nielsen, L. P. Pekowsky, H. P. Pfeiffer, P. R. Saulson, M. West, and J. L. Willis, The PyCBC search for gravitational waves from compact binary coalescence, *Classical and Quantum Gravity* **33**, 215004 (2016), arXiv:1508.02357 [gr-qc].
- [5] C. Pankow, K. Chatziioannou, E. A. Chase, T. B. Littenberg, M. Evans, J. McIver, N. J. Cornish, C.-J. Haster, J. Kanner, V. Raymond, S. Vitale, and A. Zimmerman, Mitigation of the instrumental noise transient in gravitational-wave data surrounding gw170817, *Phys. Rev. D* **98**, 084016 (2018).
- [6] B. Steltner, M. A. Papa, and H.-B. Eggenstein, Identification and removal of non-Gaussian noise transients for gravitational-wave searches, *Phys. Rev. D* **105**, 022005 (2022), arXiv:2105.09933 [gr-qc].
- [7] V. Tiwari, M. Drago, V. Frolov, S. Klimenko, G. Mitselmakher, V. Necula, G. Prodi, V. Re, F. Salemi, G. Vedovato, and I. Yakushin, Regression of environmental noise in LIGO data, *Classical and Quantum Gravity* **32**, 165014 (2015).
- [8] J. C. Driggers, S. Vitale, A. P. Lundgren, M. Evans, K. Kawabe, S. E. Dwyer, K. Izumi, R. M. S. Schofield, A. Effler, D. Sigg, P. Fritschel, M. Drago, and et. al. (The LIGO Scientific Collaboration Instrument Science Authors), Improving astrophysical parameter estimation via offline noise subtraction for advanced ligo, *Phys. Rev. D* **99**, 042001 (2019).
- [9] M. Was, R. Gouaty, and R. Bonnand, End benches scattered light modelling and subtraction in advanced virgo, *Classical and Quantum Gravity* **38**, 075020 (2021).
- [10] N. J. Cornish and T. B. Littenberg, Bayeswave: Bayesian inference for gravitational wave bursts and instrument glitches, *Classical and Quantum Gravity* **32**, 135012 (2015).
- [11] J. D. Merritt, B. Farr, R. Hur, B. Edelman, and Z. Doctor, Transient glitch mitigation in advanced ligo data, *Phys. Rev. D* **104**, 102004 (2021).
- [12] S. D. Mohanty and M. A. T. Chowdhury, Glitch subtraction from gravitational wave data using adaptive spline fitting, *Classical and Quantum Gravity* **40**, 125001 (2023).
- [13] S. D. Mohanty and E. Fahnestock, Adaptive spline fitting with particle swarm optimization, *Computational Statistics*, 1 (2020).
- [14] D. L. Donoho and I. M. Johnstone, Adapting to unknown smoothness via wavelet shrinkage, *Journal of the American Statistical Association* **90**, 1200 (1995).
- [15] M. Zevin, S. Coughlin, S. Bahaadini, E. Besler, N. Rohani, S. Allen, M. Cabero, K. Crowston, A. K. Katsaggelos, S. L. Larson, T. K. Lee, C. Lintott, T. B. Littenberg, A. Lundgren, C. Østerlund, J. R. Smith, L. Trouille, and V. Kalogera, Gravity spy: integrating advanced LIGO detector characterization, machine learning, and citizen science, *Classical and Quantum Gravity* **34**, 064003 (2017).
- [16] S. Bahaadini, V. Noroozi, N. Rohani, S. Coughlin, M. Zevin, J. Smith, V. Kalogera, and A. Katsaggelos, Machine learning for gravity spy: Glitch classification and dataset, *Information Sciences* **444**, 172 (2018).
- [17] F. Robinet, N. Arnaud, N. Leroy, A. Lundgren, D. Macleod, and J. McIver, Omicron: A tool to characterize transient noise in gravitational-wave detectors, *SoftwareX* **12**, 100620 (2020), arXiv:2007.11374 [astro-ph.IM].
- [18] S. Coughlin, S. Bahaadini, N. Rohani, M. Zevin, O. Patane, M. Harandi, C. Jackson, V. Noroozi, S. Allen, J. Areeda, M. Coughlin, P. Ruiz, C. P. L. Berry, K. Crowston, A. K. Katsaggelos, A. Lundgren, C. Østerlund, J. R. Smith, L. Trouille, and V. Kalogera, Classifying the unknown: Discovering novel gravitational-wave detector glitches using similarity learning, *Phys. Rev. D* **99**, 082002 (2019), arXiv:1903.04058 [astro-ph.IM].
- [19] S. Soni, C. P. L. Berry, S. B. Coughlin, M. Harandi, C. B. Jackson, K. Crowston, C. Østerlund, O. Patane, A. K. Katsaggelos, L. Trouille, V.-G. Baranowski, W. F. Domainko, K. Kaminski, M. A. L. Rodriguez, U. Marciniak, P. Nauta, G. Niklasch, R. R. Rote, B. Téglás, C. Unsworth, and C. Zhang, Discovering features in gravitational-wave data through detector characterization, citizen science and machine learning, *Classical and Quantum Gravity* **38**, 195016 (2021), arXiv:2103.12104 [gr-qc].
- [20] T. A. Ferreira and G. González, Using t-SNE for characterizing glitches in LIGO detectors, *Classical and Quantum Gravity* **42**, 105010 (2025), arXiv:2412.16796 [astro-ph.IM].
- [21] P. Laguarta, R. van der Laag, M. Lopez, T. Dooney, A. L. Miller, S. Schmidt, M. Cavaglia, S. Caudill, K. Driessens, J. Karel, R. Lenders, and C. Van Den Broeck, Detection of anomalies amongst LIGO's glitch populations with autoencoders, *Classical and Quantum Gravity* **41**, 055004 (2024), arXiv:2310.03453 [astro-ph.IM].
- [22] J. Kennington and Z. Yarbrough, Fisher Information Velocity: A New Geometric Channel for Precision Glitch Identification in Gravitational-Wave Detectors, *arXiv e-prints*, arXiv:2605.21531 (2026), arXiv:2605.21531 [astro-ph.IM].
- [23] C. Leys, C. Ley, O. Klein, P. Bernard, and L. Licata, Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median, *Journal of Experimental Social Psychology* **49**, 764 (2013).
- [24] J. Singha, A. Basu, M. A. Krishnakumar, B. C. Joshi, and P. Arumugam, A real-time automated glitch detection pipeline at Ooty Radio Telescope, *Monthly Notices of the Royal Astronomical Society* **505**, 5488 (2021), arXiv:2106.01942 [astro-ph.IM].

- [25] S. Klimenko, G. Vedovato, M. Drago, F. Salemi, V. Tiwari, G. A. Prodi, C. Lazzaro, K. Ackley, S. Tiwari, C. F. Da Silva, and G. Mitselmakher, Method for detection and reconstruction of gravitational wave transients with networks of advanced detectors, *Phys. Rev. D* **93**, 042004 (2016), arXiv:1511.05999 [gr-qc].
- [26] A. Longo, G. Demasi, F. Di Renzo, S. Bianchi, G. Valdes, N. Arnaud, G. Inguglia, A. Cicone, M. Lenti, F. Bucci, G. Settanta, M. Montani, and R. Cavassi, Adaptive scattered light noise subtraction in gw detectors, *Classical and Quantum Gravity* **43**, 145002 (2026).
- [27] F. Robinet, N. Arnaud, N. Leroy, A. Lundgren, D. Macleod, and J. McIver, Omicron: A tool to characterize transient noise in gravitational-wave detectors, *SoftwareX* **12**, 100620 (2020).
- [28] R. Udall, S. Hourihane, S. Miller, D. Davis, K. Chatziioannou, M. Isi, and H. Deshong, Antialigned spin of GW191109: Glitch mitigation and its implications, *Phys. Rev. D* **111**, 024046 (2025), arXiv:2409.03912 [gr-qc].
- [29] D. Davis, T. Littenberg, I. Romero-Shaw, M. Millhouse, J. McIver, F. Di Renzo, and G. Ashton, Subtracting glitches from gravitational-wave detector data during the third observing run, arXiv preprint arXiv:2207.03429 (2022).
- [30] C. de Boor, *A Practical Guide to Splines (Applied Mathematical Sciences)* (Springer, 2001).
- [31] S. D. Mohanty, Particle swarm optimization and regression analysis I, *Astronomical Review* **7**, 29 (2012).
- [32] The MathWorks, Inc., spectrogram: Spectrogram using short-time Fourier transform (), MATLAB Signal Processing Toolbox.
- [33] The MathWorks, Inc., Spectrogram Computation with Signal Processing Toolbox ().
- [34] B. Boashash, ed., *Time-Frequency Signal Analysis and Processing: A Comprehensive Reference*, 2nd ed., EURASIP and Academic Press Series in Signal and Image Processing (Academic Press, Amsterdam and Boston, 2016).
- [35] D. Davis, T. Massinger, A. Lundgren, J. C. Driggers, A. L. Urban, and L. Nuttall, Improving the sensitivity of advanced LIGO using noise subtraction, *Classical and Quantum Gravity* **36**, 055011 (2019).
- [36] R. Ormiston, T. Nguyen, M. Coughlin, R. X. Adhikari, and E. Katsavounidis, Noise reduction in gravitational-wave data via deep learning, *Phys. Rev. Research* **2**, 033066 (2020).
- [37] T. Dooney, H. Narola, S. Bromuri, R. L. Curier, C. Van Den Broeck, S. Caudill, and D. S. Tan, Time-domain reconstruction of signals and glitches in gravitational wave data with deep learning, *Phys. Rev. D* **112**, 044022 (2025), arXiv:2501.18423 [gr-qc].
- [38] H. Narola, T. Wouters, L. Negri, M. Lopez, T. Dooney, F. Cireddu, M. Wils, I. C. F. Wong, P. T. H. Pang, J. Janquart, A. Samajdar, C. Van Den Broeck, and T. G. F. Li, Null-stream-based third-generation-ready glitch mitigation for gravitational wave measurements, *Phys. Rev. D* **112**, 024079 (2025), arXiv:2411.15506 [gr-qc].
- [39] M. A. T. Chowdhury, Advancements in glitch subtraction systems for enhancing gravitational wave data analysis: A brief review (2024), arXiv:2406.01318 [gr-qc].
- [40] C. de Boor, On calculating with b-splines, *Journal of Approximation Theory* **6**, 50 (1972).
- [41] S. Wold, Spline functions in data analysis, *Technometrics* **16**, 1 (1974).
- [42] H. G. Burchard, Splines (with optimal knots) are better, *Applicable Analysis* **3**, 309 (1974).
- [43] D. L. Jupp, Approximation to data by splines with free knots, *SIAM Journal on Numerical Analysis* **15**, 328 (1978).
- [44] Z. Luo and G. Wahba, Hybrid adaptive splines, *Journal of the American Statistical Association* **92**, 107 (1997).
- [45] J. Kennedy and R. C. Eberhart, Particle swarm optimization, in *Proceedings of the IEEE International Conference on Neural Networks: Perth, WA, Australia*, Vol. 4 (IEEE, 1995) p. 1942.
- [46] H. Akaike, Information theory and an extension of the maximum likelihood principle, in *Selected Papers of Hirotugu Akaike* (Springer, 1998) pp. 199–213.
- [47] X. Huo, M. Duncan, O. Levi, J. Buckheit, S. Chen, D. Donoho, and I. Johnstone, *About WaveLab*, Tech. Rep. (Stanford University, 2000).
- [48] J. Glanzer, S. Banagari, S. Coughlin, M. Zevin, S. Baahadini, N. Rohani, S. Allen, C. Berry, K. Crowston, M. Harandi, C. Jackson, V. Kalogera, A. Katsaggelos, V. Noroozi, C. Osterlund, O. Patane, J. Smith, S. Soni, and L. Trouille, Gravity spy machine learning classifications of ligo glitches from observing runs o1, o2, o3a, and o3b, 10.5281/zenodo.5649212 (2021).
- [49] R. Abbott, T. D. Abbott, S. Abraham, F. Acernese, K. Ackley, C. Adams, R. X. Adhikari, V. B. Adya, C. Affeldt, M. Agathos, K. Agatsuma, N. Aggarwal, O. D. Aguiar, A. Aich, L. Aiello, A. Ain, P. Ajith, G. Allen, A. Allocca, P. A. Altin, A. Amato, S. Anand, A. Ananyeva, S. B. Anderson, W. G. Anderson, S. V. Angelova, S. Ansoldi, S. Antier, S. Appert, K. Arai, M. C. Araya, J. S. Areeda, M. Arène, N. Arnaud, S. M. Aronson, K. G. Arun, S. Ascenzi, G. Ashton, S. M. Aston, P. Astone, F. Aubin, P. Aufmuth, K. AultONeal, C. Austin, V. Avendano, S. Babak, P. Bacon, F. Badaracco, M. K. M. Bader, S. Bae, A. M. Baer, J. Baird, F. Baldaccini, G. Ballardini, S. W. Ballmer, A. Bals, A. Balsamo, G. Baltus, S. Banagiri, D. Bankar, R. S. Bankar, J. C. Barayoga, C. Barbieri, B. C. Barish, D. Barker, K. Barkett, P. Barneo, F. Barone, B. Barr, L. Barsotti, M. Barsuglia, D. Barta, J. Bartlett, I. Bartos, R. Bassiri, A. Basti, M. Bawaj, J. C. Bayley, M. Bazzan, B. Bécsy, M. Bejger, I. Belahcene, A. S. Bell, D. Beniwal, M. G. Benjamin, J. D. Bentley, F. Bergamin, B. K. Berger, G. Bergmann, S. Bernuzzi, C. P. L. Berry, D. Bersanetti, A. Bertolini, J. Betzwieser, R. Bhandare, A. V. Bhandari, J. Bidler, E. Biggs, I. A. Bilenko, G. Billingsley, R. Birney, O. Birnholtz, S. Biscans, M. Bisch, S. Biscoveanu, A. Bisht, G. Bissenbayeva, M. Bitossi, M. A. Bizouard, J. K. Blackburn, J. Blackman, C. D. Blair, D. G. Blair, R. M. Blair, F. Bobba, N. Bode, M. Boer, Y. Boetzel, G. Bogaert, F. Bondu, E. Bonilla, R. Bonnand, P. Booker, B. A. Boom, R. Bork, V. Boschi, S. Bose, V. Bossilkov, J. Bosveld, Y. Bouffanais, A. Bozzi, C. Bradaschia, P. R. Brady, A. Bramley, M. Branchesi, J. E. Brau, M. Breschi, T. Briant, J. H. Briggs, F. Brighenti, A. Brillet, M. Brinkmann, P. Brockill, A. F. Brooks, J. Brooks, D. D. Brown, S. Brunett, G. Bruno, R. Bruntz, A. Buikema, T. Bulik, H. J. Bulten, A. Buonanno, D. Buskulic, R. L. Byer, M. Cabero, L. Cadonati, G. Cagnoli, C. Cahillane, J. C. Bustillo, J. D. Callaghan, T. A. Callister, E. Calloni, J. B. Camp, M. Canepa, K. C. Cannon, H. Cao, J. Cao, G. Carapella, F. Carbognani,

- S. Caride, M. F. Carney, G. Carullo, J. C. Diaz, C. Casentini, J. Castañeda, S. Caudill, M. Cavaglià, F. Cavalier, R. Cavalieri, G. Cella, P. Cerdá-Durán, E. Cesarini, O. Chaibi, K. Chakravarti, C. Chan, M. Chan, S. Chao, P. Charlton, E. A. Chase, E. Chassande-Mottin, D. Chatterjee, M. Chaturvedi, H. Y. Chen, X. Chen, Y. Chen, H.-P. Cheng, C. K. Cheong, H. Y. Chia, and F. Chianini, Open data from the first and second observing runs of Advanced LIGO and Advanced Virgo, *SoftwareX* **13**, 100658 (2021), arXiv:1912.11716 [gr-qc].
- [50] R. Abbott, T. D. Abbott, F. Acernese, K. Ackley, C. Adams, N. Adhikari, R. X. Adhikari, V. B. Adya, C. Affeldt, D. Agarwal, M. Agathos, K. Agatsuma, N. Aggarwal, O. D. Aguiar, L. Aiello, A. Ain, P. Ajith, S. Akcay, T. Akutsu, S. Albanesi, A. Allocca, P. A. Altin, A. Amato, C. Anand, S. Anand, A. Ananyeva, S. B. Anderson, W. G. Anderson, M. Ando, T. Andrade, N. Andres, T. Andrić, S. V. Angelova, S. Ansoldi, J. M. Antelis, S. Antier, S. Appert, K. Arai, K. Arai, Y. Arai, S. Araki, A. Araya, M. C. Araya, J. S. Areeda, M. Arène, N. Aritomi, N. Arnaud, M. Arogeti, S. M. Aronson, K. G. Arun, H. Asada, Y. Asali, G. Ashton, Y. Aso, M. Assiduo, S. M. Aston, P. Astone, F. Aubin, C. Austin, S. Babak, F. Badaracco, M. K. M. Bader, C. Badger, S. Bae, Y. Bae, A. M. Baer, S. Bagnasco, Y. Bai, L. Baiotti, J. Baird, R. Bajpai, M. Ball, G. Ballardin, S. W. Ballmer, A. Balsamo, G. Baltus, S. Banagiri, D. Bankar, J. C. Barayoga, C. Barbieri, B. C. Barish, D. Barker, P. Barneo, F. Barone, B. Barr, L. Barsotti, M. Barsuglia, D. Barta, J. Bartlett, M. A. Barton, I. Bartos, R. Bassiri, A. Basti, M. Bawaj, J. C. Bayley, A. C. Baylor, M. Bazzan, B. Bécsy, V. M. Bedakihalé, M. Bejger, I. Belahcene, V. Benedetto, D. Beniwal, T. F. Bennett, J. D. Bentley, M. Benyaala, F. Bergamin, B. K. Berger, S. Bernuzzi, C. P. L. Berry, D. Bersanetti, A. Bertolini, J. Betzwieser, D. Beveridge, R. Bhandare, U. Bhardwaj, D. Bhattacharjee, S. Bhaumik, I. A. Bilenko, G. Billingsley, S. Bini, R. Birney, O. Birnholtz, S. Biscans, M. Bisch, S. Biscoveanu, A. Bisht, B. Biswas, M. Bitossi, M.-A. Bizouard, J. K. Blackburn, C. D. Blair, D. G. Blair, R. M. Blair, F. Bobba, N. Bode, M. Boer, G. Bogaert, M. Boldrini, L. D. Bonavena, F. Bondu, E. Bonilla, R. Bonnand, P. Booker, B. A. Boom, R. Bork, V. Boschi, N. Bose, S. Bose, V. Bossilkov, V. Boudart, Y. Bouffanais, A. Bozzi, C. Bradaschia, P. R. Brady, A. Bramley, A. Branch, M. Branchesi, J. Brandt, J. E. Brau, M. Breschi, T. Briant, J. H. Briggs, A. Brillet, M. Brinkmann, P. Brockill, A. F. Brooks, J. Brooks, D. D. Brown, S. Brunett, G. Bruno, R. Bruntz, J. Bryant, T. Bulik, H. J. Bulten, A. Buonanno, R. Buscicchio, D. Buskulic, C. Buy, R. L. Byer, G. S. C. Davies, L. Cadonati, G. Cagnoli, C. Cahillane, J. C. Bustillo, J. D. Callaghan, T. A. Callister, E. Calloni, J. Cameron, J. B. Camp, M. Canepa, S. Canevarolo, M. Cannavacciuolo, K. C. Cannon, H. Cao, Z. Cao, E. Capocasa, E. Capote, G. Carapella, and F. Carbognani, GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run, *Physical Review X* **13**, 041039 (2023), arXiv:2111.03606 [gr-qc].
- [51] M. Vallisneri, J. Kanner, R. Williams, A. Weinstein, and B. Stephens, The LIGO Open Science Center, in *Journal of Physics Conference Series*, Journal of Physics Conference Series, Vol. 610 (IOP, 2015) p. 012021, arXiv:1410.4839 [gr-qc].
- [52] The MathWorks, Inc., MATLAB Wavelet Toolbox ().
- [53] J. M. Lilly and S. C. Olhede, Generalized Morse Wavelets as a Superfamily of Analytic Wavelets, *IEEE Transactions on Signal Processing* **60**, 6036 (2012), arXiv:1203.3380 [stat.ME].
- [54] S. C. Olhede and A. T. Walden, Generalized Morse Wavelets, *IEEE Transactions on Signal Processing* **50**, 2661 (2002).
- [55] S. Chatterji, L. Blackburn, G. Martin, and E. Katsavounidis, Multiresolution techniques for the detection of gravitational-wave bursts, *Classical and Quantum Gravity* **21**, S1809 (2004).
- [56] S. Wang, *Characterization of Hardware Injections in LIGO Data*, LIGO SURF Final Report (LIGO Laboratory, California Institute of Technology, 2014) advisors: Alan Weinstein and Jonah Kanner.
- [57] D. Gerosa and M. Bellotti, Quick recipes for gravitational-wave selection effects, *Classical and Quantum Gravity* **41**, 125002 (2024), arXiv:2404.16930 [astro-ph.HE].
- [58] S. Hourihane, K. Chatziioannou, M. Wijngaarden, D. Davis, T. Littenberg, and N. Cornish, Accurate modeling and mitigation of overlapping signals and glitches in gravitational-wave data, *Phys. Rev. D* **106**, 042006 (2022).
- [59] S. Ghonge, J. Brandt, J. M. Sullivan, M. Millhouse, K. Chatziioannou, J. A. Clark, T. Littenberg, N. Cornish, S. Hourihane, and L. Cadonati, Assessing and mitigating the impact of glitches on gravitational-wave parameter estimation: A model agnostic approach, *Phys. Rev. D* **110**, 122002 (2024), arXiv:2311.09159 [gr-qc].